

HARINATH RAJU MUPPALA

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PROFESSIONAL SUMMARY

Physical Design Engineer with end-to-end RTL-to-GDSII / netlist-to-GDS2 implementation experience across SoC partition-level PnR, timing closure, power-grid integrity, and signoff verification using Synopsys and Siemens EDA tools. Demonstrated ability to achieve DRC/LVS-clean, timing-closed designs across MMMC corners in 32 nm CMOS, including hard macro integration, schematic-to-layout verification, and EMIR-aware power distribution. Seeking to contribute to cell, block-level, and full-chip physical design.

CORE COMPETENCIES

Physical Design & PnR: Partition-Level P&R · Floorplanning & Abstract Planning · Auto Place-and-Route · Clock Distribution & CTS · Power-Grid Design · Placement Optimization · Route & ECO · Full-Chip Assembly

Signoff & Verification: Static Timing Analysis (STA / MMMC) · Schematic-to-Layout (LVS) Verification · DRC / ERC · Physical Verification · Timing Closure

EDA Tools: Synopsys ICC2 · Fusion Compiler · PrimeTime · Design Compiler · Siemens Calibre · Xilinx Vivado

HDL & Scripting: Verilog · SystemVerilog · Tcl (flow automation & constraint scripting)

Concepts: SoC Back-End Design Flow · Physical Synthesis · Power-Grid Integrity · Low-Power Design · FSMs · Combinational & Sequential Logic

PROFESSIONAL TRAINING

Advanced VLSI Physical Design & Verification Trainee

Jun 2025 – Dec 2025

Maven Silicon Softech Pvt. Ltd. · Bangalore, India

- Executed full RTL-to-GDSII backend flow on industry-scale SoC designs using Synopsys ICC2 and Siemens Calibre, covering floorplanning through timing closure and signoff.
- Performed partition-level placement & route (PnR), power-grid design, and CTS under multi-mode multi-corner (MMMC) constraints
- Performed static timing analysis (STA) signoff using PrimeTime across MMMC corners, resolving setup/hold violations and closing timing with zero critical paths.
- Automated PD flows, SDC constraint management, and ECO execution with Tcl scripts, contributing to team BKMs and design guidelines.

KEY PROJECTS

SoC Physical Design – RV32I RISC-V Processor Core

Nov 2025 – Dec 2025

- Led end-to-end partition-level PnR for a RISC-V 32I core (10,919 cells, 0.0847 mm²) in 32 nm CMOS, integrating two custom hard macros via Synopsys Library Compiler.
- Achieved clean timing closure at 250 MHz across all MMMC corners with zero DRC/LVS violations in Calibre signoff, delivering a tapeout-ready GDSII.
- Designed the CTS network and a multi-power-domain grid with level shifters.
- Resolved timing violations via iterative ECO and placement-aware optimization, mirroring Intel's debug-and-reliability-verification workflow.

Router 1x3 – SoC Partition Physical Design

Oct 2025 – Nov 2025

- Completed netlist-to-GDS2 PnR for a 1x3 router IP at 500 MHz, taking the design from synthesis through placement and multi-corner timing signoff, delivering clean timing and DRC-clean signoff.
- Performed ECO-based timing fixes across process corners and refined the power grid to eliminate IR-drop violations.
- Managed partition architecture (floorplan aspect ratio, macro placement, pin assignment) in coordination with logic design intent.

8-bit Signed ALU – RTL Design

Jul 2025 – Aug 2025

- Designed and verified an 8-bit signed ALU in Verilog, covering arithmetic, logical, and shift operations with full status-flag logic.
- Analyzed Design Compiler synthesis reports to evaluate timing, area, and power trade-offs.

Asynchronous FIFO – Clock Domain Crossing Design

Jul 2025 – Aug 2025

- Designed an async FIFO with independent read/write clocks, using Gray-code pointer synchronization and accurate full/empty flag generation.

EDUCATION

B.Tech — Electronics & Communication Engineering

Nov 2021 – May 2025

RGUKT Nuzvid (Rajiv Gandhi University of Knowledge Technologies) · Nuzvid, India · CGPA: 8.15 / 10

CERTIFICATIONS

- Advanced VLSI Physical Design & Verification — Maven Silicon Softech Pvt. Ltd., 2025
- VLSI for Beginners — NIELIT, NIT Calicut, 2025
- Electronic Systems Design: Circuits & PCB Design with CAD Software — NPTEL (Elite + Silver), 2024